

VIKRAM DIXIT K

Senior Data Scientist — Predictive Modeling, Applied ML & Large-Scale Data Engineering

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Professional Summary

- Data Scientist with **15+ years** of experience spanning production data engineering and applied ML — **7.5 years** building large-scale distributed data infrastructure at Hortonworks/Cisco and **8 years** designing predictive models, ML pipelines, and experiments in applied research roles.
- Designed, launched, and evaluated a **live experiment on 40,000+ users** — authored the pre-analysis plan, defined compensable metrics, ran power calculations, and designed stratified randomization through go-live; partnered directly with CEO, Product, and Engineering leads at every stage.
- Built and shipped production ML systems: neural recommender systems (PyTorch/TensorFlow), LLM/GenAI pipelines processing **1.4M+ documents**, and predictive causal inference models — with offline evaluation, feedback loops, and safety protocols.
- Comfortable translating complex model outputs into decision-ready insights for Finance, Product, Engineering, and C-suite stakeholders; track record of educating non-technical audiences on model mechanics and data inputs.

Experience

Postdoctoral Scholar — Data Engineering & NLP

Mar 2026 – Present

Boston University, Global Development Policy Center

Mountain View, CA (Remote)

- Own and maintain a bilingual (Chinese/English) NLP data pipeline (fine-tuned BERT) ingesting sentiment signals from **1.4M+ news articles on the Chinese economy**; find heterogeneous financial market responses — a 1 s.d. shock to English (global) media drives a **+3% rise in the Shanghai Composite** and **0.5% monetary contraction**, while an equivalent Chinese (local) media shock drives a **+1% equity gain** and **1% monetary contraction**.
- Designing and delivering an **AI workshop for researchers** demonstrating how LLMs can be integrated into the research workflow — covering automated data extraction, structured output generation, and LLM-assisted analysis pipelines.

Postdoctoral Research Fellow — Applied ML & Experimentation

Jun 2024 – Oct 2025

Stanford University, Golub Capital Social Impact Lab

Stanford, CA

- **Designed and launched a large-scale experiment** on 40,000+ users of **Freadom** — an English literacy app deployed in schools across India — authored pre-analysis plan, obtained IRB approval, ran power calculations, and designed stratified randomization through go-live.
- Built and shipped **neural recommender systems** (PyTorch/TensorFlow) and a GenAI content generation pipeline with human-in-the-loop safety evaluation; benchmarked offline metrics to validate model quality.
- **Embedded with the product team**: partnered directly with the **CEO, Product, and Engineering leads** on experiment design, privacy protocols, and deployment readiness — functioning as the data science lead for a platform serving 40,000+ users.
- Mentored Master's students and pre-doctoral researchers on causal inference methodology and ML pipelines.

Doctoral Researcher — Structural Macroeconomics & Quantitative Research

Sep 2018 – May 2024

Boston University

Boston, MA

- Modeled firm-level risk propagation across a **20-year panel of 30,000+ companies** using time series and panel data methods; built equity factor portfolios that outperformed the Fama-French 5-factor benchmark (**+7% adjusted R-squared**); presented findings at the **Green Line Macro Meeting (BU/Boston College, 2023)**.
- Built research data pipelines integrating large-scale datasets spanning **patent filings, M&A transactions, and earnings sentiment** across 30,000+ firms; applied structural estimation (GMM/SMM) to quantify how financial frictions distort investment behavior and suppress productivity; served as peer reviewer for economics journals.

Senior Software Engineer — Big Data Analytics

Jan 2012 – May 2016

Hortonworks Inc. (Cloudera)

Santa Clara, CA

- Architected large-scale SQL analytics infrastructure (**Apache Hive, Spark, Tez**) for Fortune 500 clients across on-premises and **Azure** environments; led query execution optimizations that **doubled parallel Hive throughput** for a key Azure client and improved overall processing efficiency by **40%**.
- Served as **Release Manager for Apache Hive** and committer in the open-source community; coordinated cross-organization releases across contributors from multiple companies.
- Mentored junior engineers on the query execution team; reviewed designs and shipped production Java code for distributed systems components.

Software Engineer III — Computer Networks

Jan 2011 – Dec 2011

Cisco Systems Inc.

San Jose, CA

- Implemented C/C++ applications for real-time network data and Quality-of-Service (QoS) optimization.

Software Engineer I — Computer Networks

Jul 2007 – Jul 2009

Cisco Systems Inc.

Bengaluru, India

- Implemented C/C++ applications for session-based configuration management with atomicity guarantees.
- Won **Cisco Cool Ideas competition** for designing a low energy switching system.

Education

Boston University PhD in Economics	2024
<i>Dissertation: Essays on the Economic Impacts of Perceptions of Risk, Sentiment and M&A</i>	
University of Texas at Austin MA in Economics	2017
University of Southern California MS in Computer Science	2010
BMS College of Engineering Bachelor of Engineering	2007

Skills

Analytics & Statistical Modeling: Metric Design, A/B Testing, Power Analysis, Classification, Regression, Forecasting, Causal Inference (DiD, IV, TWFE, Propensity Scores), Structural Estimation (GMM/SMM), Scenario Modeling

Python/R & Workflow: Python (Pandas, NumPy, scikit-learn), R (tidyverse, fixest), Stata, Jupyter, Google Colab, RMarkdown, Quarto, Linux/Unix, Git, REST APIs

SQL & Data Platforms: SQL, Hive, Hadoop, Spark, AWS, Azure, Distributed Systems, Large-scale Data Pipeline Ownership, Data Quality & Integration

ML & NLP: Recommender Systems, PyTorch, TensorFlow, Hugging Face, LLM/GenAI Pipelines, Fine-tuned BERT, Multilingual NLP (Chinese/English), Offline Evaluation, Feedback Loop Design

Leadership & Collaboration: Cross-functional Partnership (Finance, Product, Engineering, C-suite), Stakeholder Education, Mentoring, Open-Source Community (Apache Hive Release Manager)

Data Storytelling: Stakeholder Presentations, Executive Summaries, Dashboards